

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0681

Krumkill 100 kV
March 19, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0681. The Interconnection Customer proposes a 218.2 MW solar photovoltaic facility interconnecting at the Krumkill 100 kV substation in the Illinois area. The request was received on September 26, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 280 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,344,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Davey Town Substation to Elk Grove 100.0 kV	Potter County to Prospect Hts 100.0 kV line	139.5	112	28.6
Guayama Pds to Davey Town 100.0 kV	Guayamapds Substation to Davey Town 100.0 kV Circuit 2	115.5	199	26.3
Guayama Pds to Daey Town 100.0 kV	Guayama Pds to Davey Town 100.0 kV Ckt 1	115.5	199	26.3
Prospect Heights to Potter County 100.0 kV L	Plains to Elk Grove 100.0 kV	112.8	203	37.1
Prospect Hts to Crown 100.0 kV L	Jepsen to Prospect Hts 100.0 kV	109.1	111	13.5
East to Potter County 100.0 kV	Potter County to Dell 100.0 kV	108.9	161	24.5
Crown to Sandhill 100.0 kV line	Elk Grove to Plains 100.0 kV	106.8	116	20.8
Guayama Pds to Dell 100.0 kV Line 1	Dell to Guayama Pds 100.0 kV Circuit 2	105.9	194	26.4
Guayama Pds to Dell 100.0 kV Ckt 2	Dell to Guayama Pds 100.0 kV Circuit 1	105.9	194	26.4
Dell Substation to Potter County 100.0 kV line	E to Dell 100.0 kV line	105.5	175	23.2
E to Dell 100.0 kV Line	Potter County to Dell 100.0 kV	105.2	188	24.5
Krumkill to Davey Town Substation 100.0 kV Circuit 1	Krumkill to Davey Town 100.0 kV Line 3	105.1	185	22.9
Krumkill to Daey Town 100.0 kV Cir. 2	Davey Town to Krumkill 100.0 kV #1	105.1	185	22.9
Davey Town to Krumkill Substation 100.0 kV Circuit 3	Krumkill to Davey Town 100.0 kV Ckt 1	105.1	185	22.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Elk Grove to Davey Town 100.0 kV	Rebuild line with higher-capacity conductor	\$2,830,000	57% (\$1,604,000)
Guayama Pds to Davey Town 100.0 kV	Rebuild line with higher-capacity conductor	\$1,764,000	56% (\$997,000)
Guayama Pds to Daey Town 100.0 kV	Rebuild line with higher-capacity conductor	\$1,267,000	56% (\$716,000)
Prospect Heights to Potter County 100.0 kV L	Rebuild line with higher-capacity conductor	\$1,995,000	56% (\$1,109,000)
Prospect Hts to Crown 100.0 kV L	Reconductor line	\$508,000	55% (\$277,000)
East to Potter County 100.0 kV	Reconductor line	\$272,000	56% (\$152,000)
Crown to Sandhill 100.0 kV	Reconductor line	\$750,000	48% (\$357,000)
Guayama Pds to Dell 100.0 kV Line 1	Reconductor line	\$514,000	56% (\$291,000)
Guayama Pds to Dell 100.0 kV Ckt 2	Reconductor line	\$660,000	56% (\$373,000)
Dell Substation to Potter County 100.0 kV line	Reconductor line	\$836,000	56% (\$467,000)
E to Dell 100.0 kV Line	Reconductor line	\$559,000	56% (\$313,000)
Krumkill to Davey Town 100.0 kV Line 1	Reconductor line	\$820,000	57% (\$465,000)
Krumkill to Davey Town 100.0 kV Cir. 2	Reconductor line	\$1,120,000	57% (\$635,000)
Davey Town Substation to Krumkill 100.0 kV line	Reconductor line	\$1,037,000	57% (\$588,000)
	Total allocated to project		\$8,344,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.